

Pierre Vernimmen

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 Pierre Vernimmen |  Verpierre

Louvain-la-Neuve, Belgium

EXPERIENCE

• UCLouvain

PhD student and teaching assistant

October 2024 - Now

Louvain-la-Neuve, Belgium

- Conducted research in optimization
- Gave exercise sessions and organized projects to undergraduate and graduate students
- Supervised by [Prof. François Glineur](#)

• Inria

Visiting PhD student

February 2026 - April 2026

Paris, France

- Worked on accelerated methods within the Sierra Team
- Hosted by [Adrien Taylor](#) and [Baptiste Goujaud](#)

• UCLouvain

Tutor

September 2022 - June 2024

Louvain-la-Neuve, Belgium

- Gave exercise sessions to undergraduate students
- Courses in calculus, optimization, signals and systems and algorithmic

EDUCATION

• UCLouvain

Master in Mathematical Engineering (Optimization and Data Science)

September 2022 - June 2024

Louvain-la-Neuve, Belgium

- Grade: Magna Cum Laude

• UCLouvain

Bachelor in Engineering (Applied Mathematics and Physics)

September 2019 - June 2022

Louvain-la-Neuve, Belgium

• Institut Saint-Louis

Secondary Education (Mathematics and Science)

June 2019

Namur, Belgium

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PREPRINT, T=THESIS

- [C.1] Pierre Vernimmen & François Glineur (2024). **Convergence analysis of an inexact gradient method on smooth convex functions**. In *Proceedings of the 32nd European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning*, pp. 125-130. 2024, Brugge, Belgium.
- [C.2] Pierre Vernimmen & François Glineur (2025). **Empirical and computer-aided robustness analysis of long-step and accelerated methods in smooth convex optimization..** In *Proceedings of the 5th international conference on "Modelling, Computation and Optimization in Information Systems and Management Sciences"*, Springer (to be published in November 2025), Metz, France.
- [J.1] Pierre Vernimmen & François Glineur (2026). **Worst-case convergence analysis of relatively inexact gradient descent on smooth convex functions**. *Neurocomputing*, Vol. 660, 131883
- [T.1] Pierre Vernimmen (2024). **Tight Convergence Analysis of Exact and Inexact Gradient Methods with Constant and Silver Schedules**. Master's Thesis, UCLouvain.

SKILLS

- **Programming Languages:** Python, Julia, R, C, C++
- **Mathematical & Statistical Tools:** MatLab, Wolfram Mathematica

TEACHING EXPERIENCE

• Stochastic variance-reduced optimization algorithms and applications to federated learning <i>Graduate school in optimization: gave exercise sessions, with coding exercises</i>	October 2025
• LINMA1702: Optimization Models and Methods 1 <i>Bachelor's level course in optimization: managed project</i>	2025-26
• LINMA2471: Optimization Models and Methods 2 <i>Master's level course in optimization: gave exercise sessions and projects</i>	2024-25 and 2025-26
• LEPL1106: Signals and Systems <i>Bachelor's level course: managed tutors and homeworks</i>	2024-25
• LINMA1702: Optimization Models and Methods 1 <i>Bachelor's level course in optimization: gave exercise sessions</i>	2023-24
• LEPL1106: Signals and Systems <i>Bachelor's level course: gave exercise sessions and corrected homework</i>	2022-23 and 2023-24
• LEPL1102: Calculus 1 <i>Bachelor's level course: gave exercise sessions</i>	2022-23 and 2023-24
• LINFO1113: Numerical Algorithmic <i>Bachelor's level course: gave exercise sessions</i>	2022-23

RESEARCH TALKS

EUROPT 2025 , Southampton, UK. <i>Computer-aided analysis of relatively inexact gradient descent in smooth convex optimization</i> (29 June - 2 July)	
MCO 2025 , Metz, France. <i>Empirical and computer-aided robustness analysis of long-step and accelerated methods in smooth convex optimization</i> (4 June - 6 June)	
ESANN 2024 , Brugge, Belgium. <i>Convergence analysis of an inexact gradient method on smooth convex functions</i> (9 October - 11 October)	

HONORS AND AWARDS

• Top 5 among first year students at Belgian Mathematical Olympiad <i>La Société Belge des Professeurs de Mathématiques</i>	June 2014
• Top 5 at Belgium Youth Chess Championship	2022
• Belgian Inter-School Chess Team Champion	2018

LEADERSHIP EXPERIENCE

• Scout leader <i>Les Scouts, Belgium</i> ◦ Group leader for children and teenagers ◦ Managed activities and cash flow	September 2019 - September 2024
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ADDITIONAL INFORMATION

Languages: French (Native language), English (Fluent), Dutch (Elementary knowledge)
Interests: Chess, Sports